

# BUSS386 Problem Set 1

## Introduction to Derivatives

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**Note.** This problem set is for *self-study only* — you do not need to turn it in. Solutions will be posted on the course website.

### Problem 1 — Classification

For each contract below, (a) classify it as a *forward*, *futures*, *swap*, or *option*; (b) state who would naturally be *long* (buyer) and who would be *short* (seller); and (c) say whether it would more naturally trade on an *exchange* or *OTC*, with one short reason.

- (i) Korean Air agrees today to buy 10 million barrels of jet fuel at \$80/bbl, with delivery in December 2026.
- (ii) POSCO agrees to sell 50,000 tons of steel to a German auto plant in March 2027 at a price fixed today.
- (iii) Samsung Electronics agrees to pay 4.0% fixed and receive 3-month SOFR on a notional of \$500M for 5 years.
- (iv) A KRX-listed contract obligating the holder to settle KOSPI 200 at 350 in 2 months, marked-to-market daily with initial and variation margin.
- (v) A contract giving its holder the right — but not the obligation — to buy 100 shares of NAVER at ₩200,000 per share within the next three months, paid for with a small upfront premium.

### Problem 2 — Markets, notional vs. market value

- (a) Two banks enter a 5-year interest-rate swap with a notional principal of \$100M. Six months later, the present value of the remaining net payments is \$1.2M (in favor of one side). State the contract's *notional*, *gross market value*, and which number better reflects the true economic exposure.
- (b) According to the BIS, OTC derivatives gross notional is roughly \$700T, while gross market value is roughly \$20T. Explain why these two numbers differ by such a large factor.
- (c) For each of the following, would the contract more naturally clear through a CCP, or remain bilateral under an ISDA Master Agreement?
  - (1) A standardized 3-month KOSPI 200 futures contract.
  - (2) A bespoke FX option tied to the closing of a specific M&A deal.
  - (3) A vanilla 5-year USD interest-rate swap between two large G-SIBs.

### Problem 3 — Returns and volatility

A stock closes at the following prices on 5 consecutive trading days:

|       |       |       |      |       |       |
|-------|-------|-------|------|-------|-------|
| Day   | 0     | 1     | 2    | 3     | 4     |
| Close | \$100 | \$102 | \$98 | \$103 | \$101 |

- (a) Compute the simple daily returns  $R_t$  for  $t = 1, \dots, 4$  (4 numbers).
- (b) Compute the log daily returns  $r_t$  for  $t = 1, \dots, 4$  (4 numbers).
- (c) Compute the cumulative return from Day 0 to Day 4 using both conventions. Which convention is easier to add over time, and why?
- (d) Compute the daily standard deviation of the log returns. Annualize it using the  $\sqrt{252}$  rule.
- (e) State one assumption that is required for the  $\sqrt{T}$  scaling rule to hold.

### Problem 4 — Discrete random variable (KOSPI 200 scenario)

You believe that next year's return on the KOSPI 200 ETF,  $R$ , can be summarized by three scenarios:

| Scenario    | Return $r_i$ | Probability $p_i$ |
|-------------|--------------|-------------------|
| Bull market | +22%         | 0.25              |
| Base case   | +7%          | 0.55              |
| Bear market | -18%         | 0.20              |

- (a) Compute the expected return  $E(R)$ .
- (b) Compute the variance and the standard deviation  $\sigma(R)$ .
- (c) You invest ₩50,000,000 in this ETF today. What is the expected wealth one year from now? What is the worst-case wealth?
- (d) Now consider a second asset, the KTB (Korean Treasury Bond) futures, with  $E(R_B) = 3\%$  and  $\sigma_B = 6\%$ . You hold the ETF and KTB in equal weights ( $w_{ETF} = w_{KTB} = 0.5$ ); the correlation is  $\rho = -0.2$ . Compute the portfolio's expected return and standard deviation.
- (e) Explain why the portfolio standard deviation in (d) is lower than the simple weighted average of the two individual standard deviations.

## Problem 5 — VaR and Expected Shortfall

You hold a \$2,000,000 position in an S&P 500 ETF. Daily log returns are approximately normal with mean zero and daily standard deviation  $\sigma_{\text{daily}} = 1.1\%$ .

- (a) Compute the parametric (Gaussian) 1-day 95% VaR. Use  $z_{0.05} = 1.645$ .
- (b) Compute the parametric 1-day 99% VaR. Use  $z_{0.01} = 2.326$ .
- (c) Compute the parametric 10-day 99% VaR using the  $\sqrt{T}$  scaling rule.
- (d) You also pull the last 250 trading days of returns for the same ETF. The 13th-worst return (i.e., the 5th-percentile return) is  $-2.4\%$ , and the average of the 13 worst is  $-3.0\%$ .
  - (1) Compute the empirical 1-day 95% VaR (in dollars).
  - (2) Compute the empirical 1-day 95% Expected Shortfall (in dollars).
- (e) Answer:
  - (i) Why does ES exceed VaR by construction?
  - (ii) In what kind of market environment would the empirical VaR in (d) likely *understate* true tail risk?
  - (iii) Give one real-world application (other than Basel bank capital) where VaR or ES is actually used.